Chia-Yuan (Scott) Chang

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Applied Scientist, Amazon Rufus — Palo Alto, CA

I am looking for full-time roles in LLM post-training.

TL;DR

- **At Amazon Rufus (LLM post-training for reasoning)**
  - I play a core role in LLM post-training for Amazon Rufus, focusing on SFT and RL for large-scale reasoning models.
  - I built the full coding RLVR recipe end-to-end for our ~100B MoE reasoning model — data curation and filtering (29K verifiable coding problems), verifier and reward design, GSPO training on 256 H200, and evaluation — lifting LiveCodeBench-v6 pass@1 from 68.6% to 75.9% (+7.3%).
  - I was one of the main developers of our first reliable SFT training and data pipeline for our ~800B MoE model, driving major gains on reasoning and instruction-following benchmarks (IFEval +8%, AIME24 +25.5%, AIME25 +25%).
  - I led in-house mid-training and was a key contributor to our long-context extension recipe for medium-sized LLMs.
- **Before and beyond current work**
  - Built RAG systems with retrieval filtering and chunk selection, including *MAIN-RAG* (ACL 2025).
  - Worked on efficient long-context LLM methods, including *Self-Extend* (ICML 2024 Spotlight).
  - Developed research on domain generalization, fairness, healthcare ML, and time-series forecasting.
- **Selected recognition**
  - CIKM 2023 Best Demo Paper Honorable Mention.
  - 2nd and 4th Place Awards at TREC CAsT 2020.

Work Experience

Amazon, Rufus, Palo Alto, CA — Jun. 2025 – Present
- *Applied Scientist*
- Led the full coding RLVR recipe for a ~100B MoE reasoning model — built a 29K verifiable-problem pool (execution-based rewards, model-based difficulty filtering, zero eval-set leakage) and trained with GSPO at large scale; LiveCodeBench-v6 pass@1 68.6% → 75.9% (+7.3%).
- Diagnosed and fixed reward-integrity failures at large scale (reward-grading starvation, truncation-driven response collapse); the response-budget fix turned a −4.4% decay into the +7.3% final gain.
- Lead and execute an in-house LLM CPT/mid-training.
- Core contributor to in-house LLM long-context extension recipe.

Amazon, Rufus, Palo Alto, CA — Nov. 2024 – Mar. 2025
- *Applied Scientist Intern*
- Researched state-of-the-art linearizing and hybrid LLM architectures.
- Proposed and implemented a new hybrid LLM architecture for internal exploration.

Visa Research, Foster City, CA — May 2024 – Aug. 2024
- *Research Intern*
- Developed a retrieval-augmented generation framework to identify and filter out unrelated chunks without model tuning.
- Explored business-facing application scenarios for RAG systems.

Texas A&M University, College Station, TX — Aug. 2021 – Present
- *Graduate Research Assistant*
- Developed model-agnostic approaches for domain generalization and robustness.
- Built fairness-aware machine learning methods for healthcare applications.
- Studied efficient training and adaptation strategies for large language models.

Academia Sinica, Taipei, Taiwan — Oct. 2019 – Dec. 2020
- *Research Assistant*

- Developed ranking algorithms for financial news recommendation, reducing traders' daily reading time by 41%.
- Won 2nd place in an information retrieval competition with a T5-based coreference and reranking framework.

Selected Publications [Google Scholar]

Only first-author/selected publications are listed below; a complete publication list is available on Google Scholar.

Current record includes ACL, KDD, 2 ICML, NAACL, NeurIPS; 8 first-author papers; 780+ citations.

[EACL2026] **FaithLM: Towards Faithful Explanations for Large Language Models**
Y.N. Chuang, G. Wang, **C.Y. Chang**, et al., and X. Hu
The 19th Conf. of the European Chapter of the Association for Computational Linguistics (EACL), 2026

[ACL2025] **MAIN-RAG: Multi-Agent Filtering Retrieval-Augmented Generation**
C.Y. Chang, Z. Jiang, et al., and N. Zou
The 63rd Annual Meeting of the Association for Computational Linguistics (ACL), 2025

[KDD2025] **CODA: Temporal Domain Generalization via Concept Drift Simulator**
C.Y. Chang, Y.N. Chuang, Z. Jiang, K.H. Lai, A. Jiang, and N. Zou
The 31st ACM SIGKDD International Conference on Knowledge Discovery & Data Mining (KDD), 2025

[ICML2024] **LLM Maybe LongLM: Self-Extend LLM Context Window Without Tuning**
H. Jin, X. Han, J. Yang, Z. Jiang, Z. Liu, **C.Y. Chang**, H. Chen, and X. Hu
International Conference on Machine Learning (ICML), 2024
ICML'24 Spotlight (3.5%)

[NAACL2024] **Learning to Compress Prompt in Natural Language Formats**
Y.N. Chuang, S. Li, J. Yuan, G. Wang, K.H. Lai, L. Yu, S. Ding, **C.Y. Chang**, et al., and X. Hu
Annual Conference of the North American Chapter of the Association for Computational Linguistics (NAACL), 2024

[NeurIPS2023] **Auto-PINN: Understanding and Optimizing Physics-Informed Neural Architecture**
Y. Wang, X. Han, **C.Y. Chang**, D. Zha, U. Braga-Neto, and X. Hu
Thirty-seventh Conference on Neural Information Processing Systems (Workshop), 2023

[AAAI2021] **A Multi-step-ahead Markov Conditional Forward Model with Cube Perturbations for Extreme Weather Forecasting**
C.Y. Chang*, C.W. Lu*, and C.J. Wang
The 36th AAAI Conference on Artificial Intelligence, 2021

[TREC2020] **Query Expansion with Semantic-based Ellipsis Reduction for Conversational IR**
C.Y. Chang, et al.
The Twenty-Ninth Text REtrieval Conference (TREC), 2020
2nd place award in TREC'20

Open-Source Projects

slime^{n}, Main Contributor — 2026
- Main contributor to an open-source LLM RL training framework, enabling scalable multi-policy and student-teacher post-training workflows, https://github.com/slime-n/slime-n

LongLM, Contributor — 2023
- Adapted self-extend implementation to open-source models, https://github.com/datamllab/LongLM

Education

- *Ph.D. in Computer Science*, Texas A&M University, College Station, TX — Aug. 2021 – Expected Aug. 2025
  Advisors: Dr. Na Zou and Dr. Xia Hu
- *M.S. in Structures and Materials*, National Cheng Kung University, Tainan, Taiwan — Sep. 2013 – Jun. 2015

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